

Subeshan Cobalakirouchenin

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Summary

IT undergraduate (B.Tech, graduating April 2027) with hands-on experience building full-stack web applications, secure mobile apps, and Python-based backend systems using object-oriented design. Skilled in Flask, Node.js, Flutter, and cloud-adjacent platforms (AWS, Google Cloud), with a strong debugging and problem-solving track record across five national hackathons, including a 1st-place finish and 12–22% performance gains over baseline AI models. Seeking to build secure, well-engineered, user-centered software with Cisco's Application Software team.

Education

Sri Manakula Vinayagar Engineering College – B.Tech in Information Technology August 2023 – April 2027

- CGPA: 7.86/10
- Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, Database Management Systems, Software Engineering, Web Technologies

Amrita Vidyalayam – CBSE (HSC: 71.0% | SSLC: 75.40%) June 2020 – March 2023

Projects

AI Voice Appointment Booking System | Full-Stack Web Application Oct 2025 – Dec 2025

- Designed and built a full-stack **conversational booking system** enabling users to book, reschedule, and cancel appointments via natural-language voice interaction, integrating **Vapi** for real-time voice AI
- Developed a **Node.js/Express.js** backend with **MySQL**, building **REST APIs** to manage appointment scheduling and user data
- Automated end-to-end workflows (voice events, API calls, database updates) using **n8n** to coordinate system components and reduce manual orchestration
- Designed an accessible, **user-centered UI** with React.js and Tailwind CSS, applying UX principles for clarity in a healthcare context. **Technologies:** React.js, Tailwind CSS, Node.js, Express.js, MySQL, Vapi, n8n

Wallet – Secure Document Manager | Mobile Application Nov 2025

- Built a **secure Flutter mobile application** for managing family documents, implementing **AES-256 encryption** to protect sensitive user data and preserve **user privacy**
- Integrated **Firebase** for cloud backup and used the Provider package for state management, structuring the app with **object-oriented, modular architecture**
- Designed PDF preview and UI animations following **Material Design** principles for an intuitive user experience. **Technologies:** Flutter, Dart, Firebase, AES-256 Encryption, Provider, Material Design

Quantum-Inspired Lung Cancer Detection | Python Backend & Applied ML May 2025 – Sep 2025

- Designed a hybrid neural network (PennyLane + ResNet-18) in **Python**, structuring the model and data pipeline using **object-oriented design**
- Improved detection accuracy and mAP by **12–22%** over baseline models through class-imbalance sampling strategies and optimized training
- Built a **Flask REST API** with secure authentication, applying **secure coding practices** to protect sensitive medical data. **Technologies:** Python, PyTorch, PennyLane, Flask, MySQL, SQLAlchemy, REST API

Technical Skills

Languages: Python, Java, C, JavaScript, Dart

Web & Backend: Flask, Node.js, Express.js, REST APIs, HTML, CSS, Object-Oriented Design

Mobile & Databases: Flutter, Firebase, MySQL, SQLAlchemy

Cloud & Tools: AWS, Google Cloud, Docker, Git, Linux (Basics), VS Code

AI/ML (supporting): TensorFlow, PyTorch, PennyLane, OpenCV, NumPy, Pandas

Achievements

- **LEO Summit Hackathon 2026** – Secured **1st place** in a 24-hour national hackathon
- **AI Ignite National Hackathon** – Secured **4th place** at the national level
- **AI Agents Hackathon** – Ranked **Top 6** in agent design and development
- **Unisys UIP 16** – Pre-finalist, selected among hundreds for an innovative technology solution
- **Xyntra Hackathon** – Finalist, advanced to finals for a full-stack development solution

Certifications

- Skillrack – Java, C, DS, Python, SQL Basics
- Microsoft – Gen AI Fundamentals, AI in Azure
- IBM – ML with Python; AWS – Data Analytics
- Google Cloud – 5+ Gen AI Badges; NPTEL – Ethical Hacking